Velocity Control

Given $(V_d, \omega_d) \rightarrow$ make (V and ω) converge to (V_d, ω_d)

I Unicycle Model

States: x, y, θ Inputs: $\dot{\theta}, V$

Let $V = V_d$ & $\omega = \omega_d$

II Differential Drive

States: x, y, θ Inputs: u_r, u_l

$V = \frac{r}{2}(u_r + u_l)$ let $V_d = \frac{r}{2}(u_r + u_l)$

$\omega = \frac{r}{L}(u_r - u_l)$ let $\omega_d = \frac{r}{L}(u_r - u_l)$

known unknowns

$$\begin{bmatrix} V_d \\ \omega_d \end{bmatrix} = \begin{bmatrix} r/2 & r/2 \\ r/L & -r/L \end{bmatrix} \begin{bmatrix} u_r \\ u_l \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} u_r \\ u_l \end{bmatrix} = M^{-1} \begin{bmatrix} V_d \\ \omega_d \end{bmatrix}$$

III Bicycle Model

States: x, y, θ Inputs: v, ϕ

$\underline{v = v_d}$ $\omega = \frac{v}{L}\tan\phi \Rightarrow \omega_d = \frac{v_d}{L}\tan\phi$

$$\Rightarrow \phi = \tan^{-1}\left(\frac{L\omega_d}{v_d}\right)$$

IV Smooth Unicycle Model

States: x, y, θ, v, ω Inputs: a, α

Given a desired (v_d, ω_d), force the vehicle to converge to that velocity.

$x_1 = v \quad x_2 = \omega \qquad u_1 = a \quad u_2 = \alpha$

$$\dot{x} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} u$$

$\underline{z = x - x_d}$ → The state we want to converge to

$x_d = \begin{bmatrix} v_d \\ \omega_d \end{bmatrix}$ If $z \to 0$ then $x \to x_d$

$\dot{z} = \dot{x} - \cancel{\dot{x}_d}^{\,0} = u$ Assume v_d & ω_d are constant

↳ $\dot{v}_d = \dot{\omega}_d = 0$

$u = -kz \Rightarrow \underline{\dot{z} = -kz}$

$z = \begin{bmatrix} v - v_d \\ \omega - \omega_d \end{bmatrix}$ $\left.\begin{matrix} 1/v_{dmax}^2 \\ 1/\omega_{dmax}^2 \end{matrix}\right\}$ choose values $u = \begin{bmatrix} a \\ \alpha \end{bmatrix}$ $\left.\begin{matrix} 1/a_{dmax}^2 \\ 1/\alpha_{dmax}^2 \end{matrix}\right\}$ choose values

$$Q = \begin{bmatrix} 1/V_{dmax}^2 & 0 \\ 0 & 1/\omega_{dmax}^2 \end{bmatrix} \quad R = \begin{bmatrix} 1/a_{dmax}^2 & 0 \\ 0 & 1/\alpha_{dmax}^2 \end{bmatrix}$$

Used to solve

$$\int_0^\infty \left(z^T(\tau) Q z(\tau) + u^T(\tau) R u(\tau) \right) d\tau$$

In MATLAB: $K = lqr(A, B, Q, R)$

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad u = -Kz$$

V Smooth Differential Drive

States: $x, y, \theta, \mu_r, \mu_l$ Input: α_r, α_l

$$V = \frac{r}{2}(\mu_r + \mu_l) \qquad \dot{V} = \frac{r}{2}(\dot{\mu}_r + \dot{\mu}_l) = \frac{r}{2}(\alpha_r + \alpha_l)$$

$$\omega = \frac{r}{L}(\mu_r - \mu_l) \qquad \dot{\omega} = \frac{r}{L}(\dot{\mu}_r - \dot{\mu}_l) = \frac{r}{L}(\alpha_r - \alpha_l)$$

let $u_1 = \alpha_r$ & $u_2 = \alpha_l$

$$\dot{x} = [0]x + \begin{bmatrix} r/2 & r/2 \\ r/L & -r/L \end{bmatrix} u$$

V_d and ω_d are constant

$$z = x - x_d \Rightarrow \dot{z} = \dot{x} = [0]x + Bu$$

$u = -Kz$

$$Q = \begin{bmatrix} 1/V_{dmax}^2 & 0 \\ 0 & 1/\omega_{dmax}^2 \end{bmatrix} \quad R = \begin{bmatrix} 1/\alpha_{r\,max}^2 & 0 \\ 0 & 1/\alpha_{l\,max}^2 \end{bmatrix}$$

$u = -Kz \quad K = lqr(A, B, Q, R)$